

Tutorial 7

Subspaces, linear combinations and spans

1. Let $V = \mathbb{R}^3$, $u = (2, -4, 5)$, $v = (-1, 2, 3)$. With its usual operations of addition and scalar multiplication, V is a vector space. For each of the following statements, say whether it is true or false. Give a reason for your answer.

(a) $S = \{\lambda u + v : \lambda \in \mathbb{R}\}$ is a subspace of V .

Solution: This is not true. One reason is that S does not contain the zero vector $\mathbf{0} = (0, 0, 0)$ because the equation

$$\lambda(2, -4, 5) + (-1, 2, 3) = (0, 0, 0)$$

has no solution.

(b) $v \in \text{span}(\{u\})$.

Solution: This is not true because the equation

$$(-1, 2, 3) = \lambda(2, -4, 5)$$

has no solution.

2. In an example in lectures, we introduced the vector space $F(\mathbb{R})$ of all real-valued functions defined on $\mathbb{R}$. In the same way, we can show that $F[0, 4]$, the set of all real-valued functions defined on the interval $[0, 4]$, becomes a vector space if we define addition and scalar multiplication by

$$(f + g)(x) = f(x) + g(x) \text{ and } (\lambda f)(x) = \lambda f(x) \text{ for every } x \in [0, 4],$$

where $f, g \in F[0, 4]$ and $\lambda \in \mathbb{R}$. (See Example 2.1.9 in the notes.)

For each of the subsets of $F[0, 4]$ described in (a) to (d) below, say whether

- (i) it is closed under the usual addition in $F[0, 4]$;
- (ii) it is closed under the usual scalar multiplication in $F[0, 4]$;
- (iii) it is a subspace of $F[0, 4]$

Give reasons for your answers.

(a) The set of all continuous functions in $F[0, 4]$.

Solution: (i) Yes: the sum of two continuous functions is again a continuous function.

(ii) Yes: a constant times a continuous function is again a continuous function.
 (iii) Yes: The set of all continuous functions in $F[0, 4]$ is non-empty. Hence by Definition 2.2.5, the set of all continuous functions in $F[0, 4]$ is a subspace of $F[0, 4]$.

(b) The set of all f in $F[0, 4]$ such that $f(4) = 0$.

Solution: Yes for (i), (ii) and (iii). For any $f, g \in F[0, 4]$ such that $f(4) = 0$ and $g(4) = 0$, and any scalar λ , we have

$$(f + g)(4) = f(4) + g(4) = 0 + 0 = 0$$

and

$$(\lambda f)(4) = \lambda(f(4)) = \lambda \cdot 0 = 0.$$

(c) The set of all f in $F[0, 4]$ such that $f(0) = 4$.

Solution: No for (i), (ii) and (iii). For example, for (ii), the function f defined by $f(x) = x + 4$ is in the given set, but $3f$ is not.

(d) The set of all continuous functions f in $F[0, 4]$ such that

$$\int_0^3 f(t)dt = f(3).$$

Solution: Yes for (i), (ii), and (iii). For any $f, g \in F[0, 4]$ such that $\int_0^3 f(t)dt = f(3)$ and $\int_0^3 g(t)dt = g(3)$, we have

$$\begin{aligned} \int_0^3 (f + g)(t)dt &= \int_0^3 (f(t) + g(t)) dt \\ &= \int_0^3 f(t)dt + \int_0^3 g(t)dt \\ &= f(3) + g(3) \\ &= (f + g)(3). \end{aligned}$$

This shows that the given set is closed under addition. Do the argument for closure under scalar multiplication similarly. Note also that the set is non-empty.

3. Let $M_{2 \times 2}(\mathbb{R})$ be the vector space of 2×2 real matrices, with the usual addition and scalar multiplication. Which of the following subsets are subspaces of $M_{2 \times 2}(\mathbb{R})$? Give reasons for your answers.

- (a) $\mathcal{S}_1 = \{A \in M_{2 \times 2}(\mathbb{R}) : A \text{ is an upper triangular matrix}\}.$

[**Hint:** Recall that a square matrix $A = (a_{ij})$ is upper triangular if $a_{ij} = 0$ for all $i > j$.]

Solution: $\mathcal{S}_1$ is a subspace:

- $\mathcal{S}_1$ is non-empty (for instance, the zero matrix is upper triangular).
- If $A, B \in \mathcal{S}_1$, then $a_{ij} = 0, b_{ij} = 0$ for all $i > j$. Let $C = A + B$, then $c_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0$ for all $i > j$, and so $C = A + B$ is upper triangular and hence in $\mathcal{S}_1$.
- Check in the same way that $\mathcal{S}_1$ is closed under scalar multiplication.

- (b) $\mathcal{S}_2 = \{A \in M_{2 \times 2}(\mathbb{R}) : A^2 = \mathbf{0}\}$ where $\mathbf{0}$ denotes the 2×2 zero matrix.

Solution: $\mathcal{S}_2$ is not a subspace of $M_{2 \times 2}(\mathbb{R})$ because it is not closed under addition.

To see this, notice that $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathcal{S}_2$ (because $A^2 = \mathbf{0}$) and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \in \mathcal{S}_2$ (because $B^2 = \mathbf{0}$), but $A + B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \notin \mathcal{S}_2$ (because $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = I$, not $\mathbf{0}$).

4. Let $C = \left\{ \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}, \begin{pmatrix} -5 \\ -6 \\ -4 \end{pmatrix}, \begin{pmatrix} 0 \\ -4 \\ 4 \end{pmatrix} \right\}.$

- (a) Is $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ in the span of C in $\mathbb{R}^3$? And $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$?

Solution: We have to solve

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = a \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} + c \begin{pmatrix} -5 \\ -6 \\ -4 \end{pmatrix} + d \begin{pmatrix} 0 \\ -4 \\ 4 \end{pmatrix}$$

for $a, b, c, d \in \mathbb{R}$. Gaussian elimination gives

$$\begin{pmatrix} -1 & 3 & -5 & 0 & 1 \\ -2 & 2 & -6 & -4 & 1 \\ 0 & 4 & -4 & 4 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} -1 & 3 & -5 & 0 & 1 \\ 0 & -4 & 4 & -4 & -1 \\ 0 & 4 & -4 & 4 & 1 \end{pmatrix} \\ \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} -1 & 3 & -5 & 0 & 1 \\ 0 & -4 & 4 & -4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Since this system definitely has solutions (why?), $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is in $\text{span}(C)$.

A similar calculation, with $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ replaced by $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, gives

$$\begin{pmatrix} -1 & 3 & -5 & 0 & 1 \\ 0 & -4 & 4 & -4 & -2 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

as a row-echelon form of the augmented matrix.

The system therefore has no solutions, and so $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is not in $\text{span}(C)$.

(b) When will an arbitrary vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$ be in the span of C ?

[**Hint:** Use the same approach you used in (a). You should find that $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is in the span of C if and only if x, y and z satisfy a certain equation. Find this equation. What kind of subset of $\mathbb{R}^3$ does this equation describe? The answer to this question should enable you to picture $\text{span}(C)$, and help with the following questions.]

Solution: Use the same method as in (a), with $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ replacing $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. After

Gaussian elimination, we obtain the augmented matrix

$$\begin{pmatrix} -1 & 3 & -5 & 0 & x \\ 0 & -4 & 4 & -4 & y - 2x \\ 0 & 0 & 0 & 0 & z + y - 2x \end{pmatrix}.$$

This system has solutions iff $z + y - 2x = 0$. This means that C spans a

plane in $\mathbb{R}^3$ (the one with cartesian equation $-2x + y + z = 0$).

- (c) If possible, write one of the vectors in C as a linear combination of the other vectors in C ; if not, say why it cannot be done.

Solution: It follows, by inspection or from the result of the Gaussian elimination above, that we can write

$$\begin{pmatrix} -5 \\ -6 \\ -4 \end{pmatrix} = 2 \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$$

or

$$\begin{pmatrix} 0 \\ -4 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} + 3 \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}$$

(These are not the only possibilities, of course.)

- (d) Find, if possible, a vector $\mathbf{v} \in \mathbb{R}^3$ such that $\text{span}(C \cup \{\mathbf{v}\}) = \mathbb{R}^3$. If it is not possible, explain why not.

Solution: It is possible. There are many possibilities; $\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ is one. (Make sure you know why.)

- (e) Can you find a vector $\mathbf{u} \in \text{span}(C)$ such that $\text{span}(C \cup \{\mathbf{u}\}) = \mathbb{R}^3$? Give a reason for your answer.

Solution: No. If $\mathbf{u} \in \text{span}(C)$, then $\text{span}(C \cup \{\mathbf{u}\}) = \text{span}(C)$, and this is not $\mathbb{R}^3$.

5. Let $B = \{2x^3, 4x^3 + 3x^2, 3x^3 + 2x^2 + 2, x^3 + 4x^2 + 3x + 5\}$. Is $\text{span } B = P_3$? Give a reason for your answer.

Solution: It holds that $\text{span } B = P_3$ iff for any $a_0, a_1, a_2, a_3 \in \mathbb{R}$,

$$\lambda_1(2x^3) + \lambda_2(4x^3 + 3x^2) + \lambda_3(3x^3 + 2x^2 + 2) + \lambda_4(x^3 + 4x^2 + 3x + 5) = a_3x^3 + a_2x^2 + a_1x + a_0$$

has a solution $\lambda_1, \lambda_2, \lambda_3, \lambda_4$.

To check whether this is the case, we determine if the following system of equations has a solution $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ for any given $a_0, a_1, a_2, a_3 \in \mathbb{R}$:

$$2\lambda_1 + 4\lambda_2 + 3\lambda_3 + \lambda_4 = a_3$$

$$3\lambda_2 + 2\lambda_3 + 4\lambda_4 = a_2$$

$$3\lambda_4 = a_1$$

$$2\lambda_3 + 5\lambda_4 = a_0$$

Interchanging the last two equations, it is easy to see that the corresponding augmented matrix is a matrix in row echelon form that has no row that contains only zeros in the entries corresponding to a variable. Hence for any given $a_0, a_1, a_2, a_3 \in \mathbb{R}$, the system of equations has a solution $\lambda_1, \lambda_2, \lambda_3, \lambda_4$. Therefore, B spans P_3 .

6. Let $S = \left\{ \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} \right\}$.

- (a) Find $\text{span } S$. Do not give the definition only; find an equation describing the matrices in $\text{span } S$. Your answer should have the form

$$\text{span } S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : \dots\dots\dots \right\},$$

with an equation containing a, b, c and d in the place of the dots.

Solution: $\text{span}(S)$ is the set of all 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ for which we can find real numbers $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1 \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We therefore need to determine for which a, b, c, d the system

$$2\lambda_1 + 3\lambda_2 + \lambda_3 = a$$

$$-\lambda_1 + 2\lambda_3 = b$$

$$\lambda_2 + 3\lambda_3 = c$$

$$\lambda_1 + 2\lambda_2 = d$$

has a solution. Gaussian elimination on

$$\begin{pmatrix} 2 & 3 & 1 & a \\ -1 & 0 & 2 & b \\ 0 & 1 & 3 & c \\ 1 & 2 & 0 & d \end{pmatrix} \text{ leads to the matrix } \begin{pmatrix} 1 & 2 & 0 & d \\ 0 & 1 & 3 & c \\ 0 & 0 & -4 & b + d - 2c \\ 0 & 0 & 0 & a + b - c - d \end{pmatrix}.$$

The system will therefore be consistent if and only if $a + b - c - d = 0$.

$$\text{Hence } \text{span}(S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a + b - c - d = 0 \right\}.$$

- (b) Give, if possible, a matrix in $M_{2 \times 2}(\mathbb{R})$ which is not in $\text{span } S$.

Solution: For instance the matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

7. Let V be a vector space and S and T two subspaces of V . Which of the following subsets of V are subspaces of V ? Give reasons for your answers.

- (a) $S \cap T = \{\mathbf{x} \in V : \mathbf{x} \in S \text{ and } \mathbf{x} \in T\}$.

Solution: $S \cap T$ is closed under addition: Suppose $\mathbf{a}, \mathbf{b} \in S \cap T$. Then $\mathbf{a}, \mathbf{b} \in S$, so, since S is closed under addition, $\mathbf{a} + \mathbf{b} \in S$. Similarly, $\mathbf{a} + \mathbf{b} \in T$. So $\mathbf{a} + \mathbf{b} \in S \cap T$. A similar argument shows that $S \cap T$ is closed under scalar multiplication. Since $\mathbf{0} \in S \cap T$, $S \cap T \neq \emptyset$, and so $S \cap T$ is a subspace of V .

- (b) $S \cup T = \{\mathbf{x} \in V : \mathbf{x} \in S \text{ or } \mathbf{x} \in T\}$.

Solution: $S \cup T$ is not necessarily closed under addition. As a counter example take $V = \mathbb{R}^2$ with its usual addition and scalar multiplication, and $S = \{(x, 0) : x \in \mathbb{R}\}$ (the x -axis) $T = \{(0, y) : y \in \mathbb{R}\}$ (the y -axis). But $S \cup T$ is not closed under addition, because $(3, 0) \in S \cup T$ and $(0, 4) \in S \cup T$, but $(3, 0) + (0, 4) = (3, 4)$ and $(3, 4) \notin S \cup T$. This is enough to show that $S \cup T$ is not a subspace of V .

Finally, for those of you who want a bit of a challenge, we have the following:

8. Let S and T be subspaces of a vector space V . Prove or disprove:

- (a) $S + T = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in S, \mathbf{v} \in T\}$ is a subspace of V ,

Solution: The statement is true.

Since $\mathbf{0} \in S$ and $\mathbf{0} \in T$, $\mathbf{0} = \mathbf{0} + \mathbf{0} \in S + T$, so $S + T \neq \emptyset$.

Suppose $\mathbf{a}, \mathbf{b} \in S + T$. Then there are vectors $\mathbf{x}_1, \mathbf{x}_2 \in S, \mathbf{y}_1, \mathbf{y}_2 \in T$ such that $\mathbf{a} = \mathbf{x}_1 + \mathbf{y}_1$ and $\mathbf{b} = \mathbf{x}_2 + \mathbf{y}_2$. Hence

$$\mathbf{a} + \mathbf{b} = (\mathbf{x}_1 + \mathbf{y}_1) + (\mathbf{x}_2 + \mathbf{y}_2) = (\mathbf{x}_1 + \mathbf{x}_2) + (\mathbf{y}_1 + \mathbf{y}_2).$$

Since S is a subspace, $\mathbf{x}_1 + \mathbf{x}_2 \in S$; since T is a subspace, $\mathbf{y}_1 + \mathbf{y}_2 \in T$. Hence $\mathbf{a} + \mathbf{b} \in S + T$, i.e. $S + T$ is closed under addition.

In the same way it can be proved that $S + T$ is closed under scalar multiplication.

Hence $S + T$ is a subspace of V .

- (b) $S + T$ is the smallest subspace of V containing $S \cup T$.

Solution: The statement is true.

$S + T$ contains $S \cup T$. To see this, let $x \in S \cup T$. Then $\mathbf{x} \in S$ or $\mathbf{x} \in T$. In the first case we have $\mathbf{x} = \mathbf{x} + 0 \in S + T$; similarly if $\mathbf{x} \in T$, $\mathbf{x} = 0 + \mathbf{x} \in S + T$.

Let W be any subspace of V which contains $S \cup T$. If $\mathbf{z} \in S + T$, then $\mathbf{z} = \mathbf{x} + \mathbf{y}$, with $\mathbf{x} \in S, \mathbf{y} \in T$. Since $\mathbf{x}, \mathbf{y} \in W$ and W is a subspace, $\mathbf{z} = \mathbf{x} + \mathbf{y} \in W$. Hence $S + T \subseteq W$.